



Prediction of extreme events in multiscale simulations of geophysical turbulence using reinforcement learning

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ABSTRACT

Accurate subgrid-scale closures are essential for weather/climate models, where predicting extreme events is critical. Traditional closures have structural errors, e.g., producing excessive diffusion that dampens extremes. Artificial intelligence has gained attention for closure modeling, but the prediction of extreme events remains challenging. Supervised offline learning needs abundant high-fidelity training data and can lead to instabilities. Online learning algorithms are emerging as an alternative, but reliance on differentiable numerical solvers or scalable optimizers hinders broad use. Here, we introduce scientific multi-agent reinforcement learning (SMARL) to develop closures for canonical prototypes of atmospheric/oceanic turbulence, using only the enstrophy spectrum, estimated from a few high-fidelity samples, as reward. This reward ensures that the model captures the cascades of scales in these simulations. These online-learned closures enable stable simulations with up to five orders of magnitude coarser spatiotemporal resolution that reproduce high-fidelity simulation statistics and substantially improve the representation of extremes: they predict the frequency of extreme vorticity events to within a factor of 2.6 of the reference, whereas positively clipped dynamic closures are off by up to a factor of 60 or miss such events entirely, although the most extreme tails remain underestimated. We interpret the closures by analyzing the SMARL policy and demonstrate generalization to other flows. The results highlight SMARL as a potent tool for developing closures that substantially improve the representation of extremes in atmospheric/oceanic flows, opening new capabilities for effective climate modeling.

1. Introduction

Extreme weather events have enormous socioeconomic impacts [1, 2], increasing the demand for the accurate estimates of their statistics, e.g., frequency, especially in a changing climate [3–5]. However, due to their rarity and complex physics, high-fidelity simulations of these extreme events needs global climate models (GCMs) that are fast, thus of moderate to low resolution, while adequately representing the underlying multi-scale physics of the atmosphere, and more broadly, the Earth system. Practically fast-enough GCMs, which have typical resolutions of $\mathcal{O}(10) - \mathcal{O}(100)$ km, require sub-grid scale (SGS) closures (parameterizations) for many physical processes [6–8]. However, traditional physics-based closures have various parametric and structural errors, making them the main source of epistemic uncertainty in climate change predictions and weather forecasts, especially for extreme

events [9–12]. Artificial intelligence (AI) applications for enhancing efficiency, e.g., through direct emulation of high-resolution simulations from GCMs or other models [13–15] or rare-event sampling with AI + GCM [16], are emerging. However, these applications are in their early stages, and AI models have been shown to have challenges with the most extreme weather events, particularly those rarer and stronger than events in the training set, the so-called gray swans [16,17]. Furthermore, even with these approaches, the need for high-fidelity closures for GCMs remains [18].

Recently, there has been great interest in using AI, particularly deep neural networks (DNNs), to develop better closures by learning directly from data. These AI-based approaches can be broadly categorized into two groups: supervised (offline) learning and online learning [12]. In supervised learning, snapshots of high-fidelity data (e.g., from direct numerical simulation (DNS)) are used as reference, then filtered and

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Table 1

The test cases and hyper-parameters in training. σ_ω is standard deviation of vorticity.

Case	Re	β	k_f	σ_ω	N_{DNS}	N_{LES}	$\Delta t_{\text{RL}}/\Delta t_{\text{LES}}$	Training horizon	Testing horizon
1	20×10^3	0	4	5.76	1024	32	10	$10^3 \Delta t_{\text{RL}}$	$10^6 \Delta t_{\text{RL}}$
2	20×10^3	20	4	10.17	1024	32	10	$10^3 \Delta t_{\text{RL}}$	$10^6 \Delta t_{\text{RL}}$
3	20×10^3	50	10	13.31	1024	128	10	$10^3 \Delta t_{\text{RL}}$	$10^6 \Delta t_{\text{RL}}$
4	20×10^3	0	25	13.24	1024	256	10	$10^3 \Delta t_{\text{RL}}$	$10^6 \Delta t_{\text{RL}}$
5	300×10^3	0	4	6.42	4096	32	10	$10^3 \Delta t_{\text{RL}}$	$10^6 \Delta t_{\text{RL}}$

coarse-grained to extract the SGS terms [19–21]. The filtered data are then used to train a DNN to learn the mapping between the filtered flow and the SGS terms such as fluxes. Once trained, the DNN is coupled to a low-resolution numerical solver, the large-eddy simulation (LES) hereafter, which represents a practical GCM. Studies with a variety of canonical prototypes of atmospheric and oceanic turbulence have shown the possibility of outperforming classical physics-based closures such as Smagorinsky and Leith models [20,22–25]. In particular, it was found that these DNN-based closures not only reproduce the right amount of diffusion, but also capture backscattering, i.e., the forcing of the large-scale flow by the SGS eddies, an important but challenging process for physics-based closures [7,26].

However, to work accurately, the supervised-learning approach has several major shortcomings: it is data hungry (a significant bottleneck for realistic systems), requires exact SGS terms extracted from high-fidelity data (a non-trivial task), and can lead to unstable LES (with the root cause(s) remaining unclear) [12,27,28]. As a result, supervised-learning with DNNs faces significant challenges in scaling up beyond prototypes and to GCMs. More recently, supervised-learning with physics-constrained equation-discovery has shown promises as a data-efficient tool for developing accurate, generalizable, and interpretable closures [29]. However, the LES with these closures is not stable for very low resolutions including those that could be of practical interest in atmosphere and ocean modeling.

Online learning is emerging as a potent alternative to supervised learning, with the closure being learned by a DNN while it is directly coupled to the low-resolution numerical (LES) solver. Here, the goal may not be to match spatially local flow quantities (such as the state's trajectory or the SGS field), but rather, to match the low-order statistics of the high-fidelity simulations (e.g., DNS) or even (sparse) observations. In climate modeling, these statistics could be key properties of climate variability [30] or kinetic energy spectra of turbulent flows [31, 32]. Finally, online learning can account for the interactions between several closures, a common problem in climate modeling [11] and unaddressed by supervised learning. We distinguish three related approaches to online learning: (1) using a differentiable LES solver/GCM that allows using backpropagation for optimization [33–35], (2) using derivative-free ensemble-based optimization methods; e.g., ensemble Kalman inversion (EKI) [30,36–40], and (3) using reinforcement learning [41–44]. The first two approaches are actively pursued in the climate community with promising results, including the scale up of approach (1) to realistic weather and climate prediction in Google's NeuralGCM [45]. However, several major challenges remain. For example, most current GCMs are not differentiable, and a differentiable solver requires a significant change in the modeling and software infrastructure [18]. Approach (2) works with non-differentiable solvers, however, is largely limited to addressing parametric uncertainty but not structural uncertainty [40]. Indeed, in EKI-type calibration the functional form of the closure is fixed *a priori* and only a small set of parameters is estimated; structural deficiencies of the assumed form, such as the inability of positively clipped eddy-viscosity models to represent backscatter, cannot be corrected by the calibration. In contrast, reinforcement learning can learn the full map from the resolved flow state to the closure coefficients, thereby modifying the effective structure of the closure itself.

scientific multi-agent reinforcement learning (SMARL) has improved the parametric and structural uncertainties of closures for LES of 3D

homogeneous and wall-bounded turbulent flows [41–44]. However, its potential for climate modeling applications has remained unexplored. We note that the extension of models for homogeneous and wall-bounded flows to LES of geophysical turbulence and to GCM is not straightforward. In LES of homogeneous turbulent flows [41], the states were based on specific well-known invariants of these flows. Similarly SMARL-based wall models of turbulent flows required knowledge of the law of the wall to determine effective states and actions [43].

In this paper, we introduce reinforcement learning for the closure modeling of geophysical turbulence. As summarized in Fig. 1, we apply SMARL to a canonical prototype: several setups of 2D turbulence that contain a broad range of scales, structures (jets and vortices), and dynamics, well representing some of the key characteristics of atmospheric and oceanic flows (see Table 1 for details). Details of the approach can be found in Methods. Briefly,

- i We train a SMARL-based closure using low-order statistics. We use global states and targets to train a DNN and learn the flow-dependent coefficient c_l of a physics-based closure (Leith model) by matching the enstrophy spectrum of the LES+SMARL solver with that of a very short DNS. These few training samples are far too few for DNN-based supervised learning. The LES solver does not need to be differentiable here.
- ii We test the performance of this closure by comparing the kinetic energy spectrum and vorticity probability density function (PDF) of the DNS with LES having 160 to $163,840 \times$ spatio-temporally coarser resolution (i.e., $(N_{\text{DNS}}/N_{\text{LES}})^2 = 16$ to $16,384 \times$ fewer grid points, combined with $10 \times$ larger time steps; see Table 1). The LES+SMARL spectrum and PDF match that of the DNS for simulations that are $\sim 2000 \times$ the length of the training dataset. In particular, the PDFs match at the tails, which represent the rare extreme weather events in these prototypes. The LES+SMARL work at resolutions much lower than those required by the recently AI-discovered closure of [29].
- iii We analyze the interpretability of the SMARL-based closures, using Sobol indices to indicate the significance of contributions of the states to the actions. We also show that these closures learn both diffusion and backscattering, with the distribution of c_l substantially differing from those estimated using dynamic Leith. This shows the ability of this approach in addressing both parametric and structural uncertainty.
- iv We demonstrate successful generalization of the SMARL-based closures on a turbulent flow with a Reynolds number (Re) $15 \times$ higher than its training DNS data.

Together, (i)–(iv) demonstrate the promises of SMARL over supervised learning and other online learning approaches. A complete list of the symbols and abbreviations used throughout the paper is provided in Appendix A.

2. Methods

2.1. Direct and large-eddy simulations of 2D turbulence

We consider two-dimensional flows expressed in the vorticity (ω) and stream function (ψ) formulation in a doubly periodic square domain with length $L = 2\pi$,

$$\frac{\partial \omega}{\partial t} + \mathcal{N}(\omega, \psi) = \frac{1}{\text{Re}} \nabla^2 \omega - f - \alpha \omega + \beta \frac{\partial \psi}{\partial x}, \quad (1a)$$

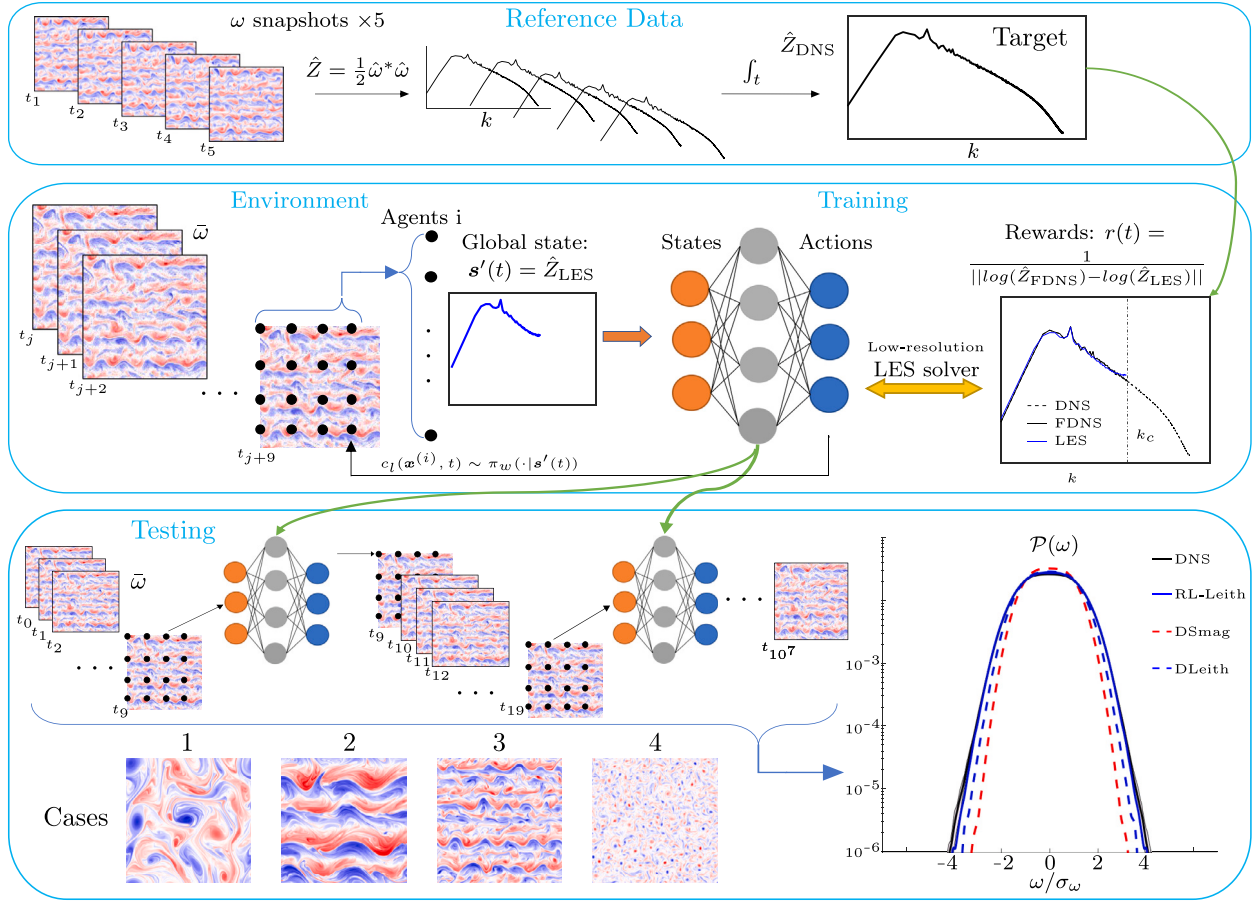


Fig. 1. Training of SMARL for SGS closures: (1) reference data consists of 5 samples of a short DNS (Eq. (1a) and (1b)); (2) online training of agents. The state-action map has input state $s'(t)$ (spectrum of entrophy $\hat{Z}$ up to LES cutoff wavenumber k_c) and output action $c_t(t)$ (coefficient in the closure, Eq. (3)) that maximizes the reward $r(t)$. During testing, the policy is coupled to the low-resolution numerical solver to produce a long LES that is 2000 $\times$ longer than the DNS training set and 1000 $\times$ the training horizon. Performance for extreme events is evaluated in terms of the vorticity PDF $\mathcal{P}(\omega)$, against DNS and dynamic physics-based SGS models.

$$\nabla^2 \psi = -\omega, \quad (1b)$$

where $\mathcal{N}(\omega, \psi) = (\partial\psi/\partial y)(\partial\omega/\partial x) - (\partial\psi/\partial x)(\partial\omega/\partial y)$, is the nonlinear advection term, and $f(x, y) = k_f[\cos(k_f x) + \cos(k_f y)]$ is a steady (time-independent), deterministic forcing [46,47], injecting energy and entrophy at the fixed wavenumber k_f . Furthermore, Re is the Reynolds number, α is the linear (Rayleigh) drag coefficient, and β is the meridional gradient of the Coriolis parameter in the β -plane approximation ($\beta \neq 0$ in Cases 2 and 3 only; see Table 1). To obtain target statistics, we solve Eq. (1) numerically. A Fourier-Fourier pseudo-spectral solver is used to solve these equations, see e.g. [20,21,25]. The computational grid has uniform spacing $\Delta_{\text{DNS}} = L/N_{\text{DNS}}$, where N_{DNS} is the number of grid points in each direction, as shown in Table 1. The time-stepping size is $\Delta t_{\text{DNS}} = 5 \times 10^{-5}$ dimensionless time unit for cases 1, 2, 3, 4 and $\Delta t_{\text{DNS}} = 1 \times 10^{-5}$ dimensionless time unit for Case 5. Once the flow reaches statistical equilibrium after a long-term spin-up, we uniformly sample 5 snapshots that are $10000\Delta t_{\text{DNS}}$ apart to obtain decorrelated data to generate the reference entrophy spectra, following [40,48].

We perform filtered direct numerical simulation (FDNS) by applying a sharp Fourier cut-off filter (cut-off wavenumber $k_c = \pi/\Delta_{\text{FDNS}}$) to each variable in Eq. (1), denoted by $\bar{\cdot}$, along each of x and y directions [49, 50]:

$$\frac{\partial \bar{\omega}}{\partial t} + \mathcal{N}(\bar{\omega}, \bar{\psi}) = \frac{1}{\text{Re}} \nabla^2 \bar{\omega} - \bar{f} - \alpha \bar{\omega} + \beta \frac{\partial \bar{\psi}}{\partial x} - \underbrace{(\mathcal{N}(\bar{\omega}, \bar{\psi}) - \mathcal{N}(\bar{\omega}, \bar{\psi}))}_{\Pi = \nabla \times (\nabla \cdot \bar{\tau}_{ij}^{\text{SGS}})}, \quad (2a)$$

$$\nabla^2 \bar{\psi} = -\bar{\omega}. \quad (2b)$$

The LES equation is the same as Eq. (2) except for the unclosed SGS term Π . The LES is solved on a coarse resolution $\Delta_{\text{LES}} = \Delta_{\text{FDNS}}$ with Π modeled as a map connecting the resolved flow variables to the closure term. The LES is solved with the same numerical method as the DNS, with grid spacing $\Delta_{\text{LES}} = L/N_{\text{LES}}$ and time step $\Delta t_{\text{LES}} = 10\Delta t_{\text{DNS}}$.

2.2. Online learning of closures with SMARL

Since the pioneering work of Smagorinsky [51] for the closure modeling of geophysical turbulence, various eddy-viscosity models have been proposed. However, the Smagorinsky [51] and Leith [52] models remain today the most widely used closures. In these models, energy transfer from the resolved to subgrid scales is formulated as non-linear diffusion [50], often with dynamic coefficients that depend on space and time [53,54]. The parameters of these models are estimated from local constraints, such as the Germano identity of energy transfer in dynamic Smagorinsky (DSmag), and the entrophy transfer in dynamic Leith (DLeith). The coefficients in Smagorinsky (c_s) and Leith (c_l) models are used to estimate the turbulence eddy viscosity: $\nu_e = c_s \Delta^2 |\bar{S}|$ [55], and $\nu_e = c_l \Delta^3 |\nabla \bar{\omega}|$ [52,56], respectively. Using these expressions, the turbulent shear stress, or subgrid stress, has the same form in both models [49,50,57]:

$$\tau_{ij}^{\text{SGS}} = -2\nu_e \bar{S}_{ij}, \quad \bar{S}_{ij} = \frac{1}{2} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right). \quad (3)$$

In this work, we use SMARL to learn a fully-connected DNN that estimates the Leith coefficient (c_l) dynamically (also referred to as RL-Leith) and compare its performance to DLeith and DSmag (we have

found RL-Leith and RL-Smag to perform similarly; see [Appendix E](#)). In the context of the SMARL framework, the state of the system $s'(t)$ is defined by the enstrophy spectrum $\hat{Z}(k, t)$ up to the Nyquist wavenumber at LES (or cut-off wavenumber) k_c . The RL-Leith coefficients are thus expressed as $c_l(x, y, t) = f_{\text{DNN}}(\hat{Z}(k, t))$. We place $n_{A,x} \times n_{A,y}$ agents distributed uniformly on the simulation grid. These agents produce actions that are interpolated onto the computation grid using a polynomial of 2nd-order that preserves the periodicity of the domain. See [Fig. 1](#) and the following subsections for the details of the SMARL setup, including the reward function (Eq. (4)) and the training procedure.

2.3. SMARL setup

We cast the closure problem as a partially observable Markov decision process addressed by $N = n_{A,x} \times n_{A,y}$ cooperating agents located at fixed positions $\mathbf{x}^{(i)}$, $i = 1, \dots, N$, distributed uniformly on the grid of the simulation domain. At every actuation time $t_j = j \Delta t_{\text{RL}}$, where $\Delta t_{\text{RL}} = 10 \Delta t_{\text{LES}}$ is the actuation interval, each agent observes a state $s^{(i)}(t_j) \in \mathbb{R}^{k_c}$ and samples a continuous action $a^{(i)}(t_j) \in \mathbb{R}$ from a shared stochastic policy, $a^{(i)}(t_j) \sim \pi(\cdot | s^{(i)}(t_j))$, represented by a fully-connected DNN with weights θ (Section 2.4). The actions set the closure coefficients at the agents' locations, $c_l(\mathbf{x}^{(i)}, t_j) = a^{(i)}(t_j)$, which are interpolated onto the LES grid to form the coefficient field $c_l(\mathbf{x}, t_j)$ entering the subgrid stress (Eq. (3)); the coefficients are kept constant until the next actuation time. All agents receive the common reward of Eq. (4), making the setting fully cooperative, and the shared policy is trained to maximize the expected discounted return $J(\theta) = \mathbb{E}_{\pi_\theta} [\sum_j \gamma^j r(t_j)]$, where γ is the discount factor. The procedure is summarized in Algorithm 1. In this work, we use global states only, meaning that $s^{(i)}(t) = s'(t)$ for any agent i . In [Appendix E](#), we show that adding local states to the input of the policy does not improve the results. We note that the sufficiency of global states is a property of the statistically homogeneous flows considered here (including the zonally homogeneous β -plane cases with strong jets) rather than a restriction of the framework: the agents are distributed across the domain and can sense local states and produce spatially varying actions under a shared policy, as demonstrated in SMARL closures for 3D homogeneous and wall-bounded turbulence [41,43]. For strongly inhomogeneous flows, e.g., with boundaries or localized jets, local states are expected to become necessary. The choice of using the same policy, $\pi(\cdot | s^{(i)}(t))$, for every agent is motivated by the translation invariance of the Navier–Stokes equations. The selection of states, actions, and rewards is critical to the performance of SMARL as detailed below.

States. Each agent observes the enstrophy spectrum of the LES, $s'(t) = \hat{Z}_{\text{LES}}(k, t)$. We found that adding the five Galilean-invariant quantities derived from filtered velocity gradients and Hessians [41,58] did not improve the performance of the closure model [E](#).

Action. The agents control local coefficients $c_l(x, y, t)$ in traditional eddy-viscosity closures: Leith model [52]. These coefficients are evaluated at the agents' locations and then interpolated onto the LES grid. The interpolated coefficients are then used to compute the subgrid stress tensor $\tau_{ij}^{\text{SGS}} = -2\nu_e \bar{S}_{ij}$, which enters the LES solver. Actions are evaluated every 10 time steps of the LES solver, and local coefficients are kept constant between two consecutive actions.

Reward. The objective of the agents is to learn a closure that matches the DNS enstrophy spectrum. Thus, we have chosen the instantaneous reward to be large when the enstrophy spectrum is close to the reference,

$$r(t) = \frac{1}{\|\log \hat{Z}_{\text{DNS}}(k) - \log \hat{Z}_{\text{LES}}(k, t)\|_2^2}, \quad (4)$$

where $\hat{Z}_{\text{DNS}}$ is the spectrum obtained from DNS simulations. This reference spectrum is computed from short DNS trajectories, which are known to be insufficient for purely offline training [48]. Furthermore,

these DNS trajectories are much shorter than what would be required to capture the rare events that we report in the results section. The enstrophy spectrum is a natural training target for 2D turbulence: it encodes the distribution of enstrophy across all resolved scales and, since $\hat{Z}(k) = k^2 \hat{E}(k)$, it simultaneously constrains the kinetic energy spectrum while giving more weight to the smallest resolved scales, which are the most affected by the SGS closure. The logarithm in Eq. (4) ensures that all wavenumbers contribute to the reward despite the rapid decay of the spectrum. This choice builds on the comparison of reward formulations by Novati et al. [41], who trained SMARL-based closures with two alternative rewards, one based on the local Germano-identity error and one based on a covariance-weighted log-likelihood of the LES energy spectrum with respect to the DNS statistics, and found the spectrum-based reward clearly superior, as the Germano-identity-based policy over-estimates dissipation. The reward in Eq. (4) is a simplified, unweighted form of their spectrum log-likelihood, which does not require estimating the covariance of the reference spectrum from the few available DNS snapshots; we found it to perform equally well. We emphasize that the reward does not target extreme events directly: a reward based on the tails of the vorticity PDF is not practical, as estimating the density of extremely rare events would require reference statistics that are orders of magnitude longer than the few DNS snapshots available for training. In contrast, the spectrum converges from a handful of decorrelated snapshots. Capturing the tails of the vorticity PDF is instead an emergent outcome of learning a closure with the correct interscale enstrophy transfer, including backscatter, and is verified only *a-posteriori* in the Results section.

Algorithm 1 SMARL training of the closure coefficient policy.

- 1: Compute the reference spectrum $\hat{Z}_{\text{DNS}}(k)$, $k \leq k_c$, from 5 decorrelated DNS snapshots
 - 2: Initialize the policy weights θ and the replay buffer
 - 3: **for** each training episode **do**
 - 4: Initialize the LES vorticity field from a filtered DNS snapshot
 - 5: **for** $j = 0, 1, \dots$ until the training horizon $10^3 \Delta t_{\text{RL}}$ **do**
 - 6: Observe the global state $s'(t_j) = \hat{Z}_{\text{LES}}(k, t_j)$, $k \leq k_c$
 - 7: Each agent i samples an action $a^{(i)}(t_j) \sim \pi^\theta(\cdot | s'(t_j))$
 - 8: Interpolate $\{a^{(i)}(t_j)\}_{i=1}^N$ onto the LES grid to obtain $c_l(\mathbf{x}, t_j)$; evaluate ν_e and τ^{SGS} (Eq. (3))
 - 9: Advance the LES by $\Delta t_{\text{RL}} = 10 \Delta t_{\text{LES}}$
 - 10: Compute the reward $r(t_{j+1})$ (Eq. (4)) and store the experiences in the replay buffer
 - 11: Update θ with V-RACER/ReF-ER (Section 2.4)
 - 12: **end for**
 - 13: **end for**
-

2.4. Training algorithm: V-RACER

The training of the policy is performed with the V-RACER algorithm with Remember and Forget experience replay (ReFER) [59], adapted to multi-agent settings as described in Weber et al. [60] and implemented in the software Korali [61]. Here, we describe the main components of the training algorithm.

At each time step t , agents observe the state $s'(t)$, perform an action sampled from the behavior policy $a(t) \sim \mu_t(\cdot, s'(t))$, and transition to a new state $s'(t + \Delta t)$ following the dynamics of the model (e.g., LES solver with SMARL-based closure here), with reward $r(t + \Delta t)$. Experiences $\{s'(t), a(t), s'(t + \Delta t), r(t), \mu_t\}$ are stored in a replay buffer. In V-RACER, the policy $\pi^\theta(\cdot, s)$ is represented as a Gaussian with mean $m^\theta(s)$ and covariance $\Sigma^\theta(s)$, which, in addition to the value $V^\theta(s)$, are the output of a single neural network with weights θ and input s .

The weights θ are updated to minimize two losses, corresponding to the off-policy objective and to match the value function with an on-policy value estimate. We use the Adam gradient-based optimization

method to minimize these objectives. The off-policy objective is used to update the policy weights:

$$\mathcal{L}^{\text{off-PG}}(\theta) = -\mathbb{E}_{s_k \sim B, a_k \sim \mu_k(s_k)} [\rho_k (\hat{Q}_t^{\text{ret}} - V^\theta(s_k))], \quad (5)$$

where $B(s) \propto \sum_{k=t-N}^t P(s_k = s | s_0, a_k \sim \mu_k)$ is the probability of sampling the state s from the replay memory containing the past N experiences of the agent acting with policy μ_k . Furthermore, the importance weight is defined as $\rho_k = \pi^\theta(s_k | a_k) / \mu_k(s_k | a_k)$, and the on-policy returns are estimated as

$$\hat{Q}_{t-1}^{\text{ret}} = r_t + \gamma V^\theta(s_t) + \gamma \bar{\rho}_t [\hat{Q}_t^{\text{ret}} - V^\theta(s_t)], \quad (6)$$

with $\bar{\rho}_t = \min\{1, \rho_t\}$. The on-policy state-value is estimated from V^θ using the recursive formula [59]

$$\hat{V}_t^{\text{tbc}} = V^\theta(s_t) + \bar{\rho}_t [r_{t+1} + \gamma \hat{V}_{t+1}^{\text{tbc}} - V^\theta(s_t)]. \quad (7)$$

This estimate is part of the loss used to train the value function:

$$\mathcal{L}^{\text{tbc}}(\theta) = \frac{1}{2} \mathbb{E}_{s_k \sim B, a_k \sim \mu_k(s_k)} [V^\theta(s_t) - \hat{V}_t^{\text{tbc}}]^2. \quad (8)$$

The gradients of these losses are updated according to the ReF-ER procedure [59]. This consists of clipping gradients to zero when the importance weight is outside of the interval $[1/c_{\max}, c_{\max}]$, where $c_{\max}$ is a positive constant. Furthermore, the gradients are biased towards the direction of past behaviors,

$$\mathbf{g}_t^{\text{ReF-ER}}(\theta) = \beta_{\text{RF}} \mathbf{g}_t(\theta) - (1 - \beta_{\text{RF}}) \nabla D_{\text{KL}} [\mu_t(\cdot | s_t) \| \pi^\theta(\cdot | s_t)], \quad (9)$$

where $\beta_{\text{RF}} > 0$ is a hyper-parameter of the ReF-ER algorithm and D_{KL} represents the KL-divergence between two distributions.

2.5. State-action map sensitivity analysis

To determine the effects of the SMARL DNN input ($\hat{Z}(k, t)$) at each wavenumber k to the NN output ($c_l(t) = f_{\text{DNN}}(\hat{Z}(k, t))$), we use the Sobol index which decomposes the variance of the output into contributions from: (1) each individual input, and (2) interaction among inputs. We emphasize that the sensitivity analysis is applied to the policy map f_{DNN} itself, which is a scalar-valued function shared by all agents; the spatial dependence of the coefficient field $c_l(x, y, t)$ arises only from the per-agent sampling of the stochastic policy and the subsequent interpolation. The scalar formulas below therefore apply directly. The same analysis applied to policies that additionally observe local, space- and time-dependent states is reported in Appendix E, where these local inputs are found to carry near-zero Sobol indices. The variance of c_l can be decomposed as:

$$\text{Var}(c_l) = \sum_m \xi_m + \sum_{m < n} \xi_{mn} + \dots + \xi_{1,2,3,\dots,k_c}, \quad (10)$$

where here $\xi_m = \text{Var}(\mathbb{E}(c_l | \hat{Z}(m)))$ are the partial variances associated with one component of the decomposition. The first-order Sobol index is defined as

$$S_j = \frac{\xi_j}{\text{Var}(c_l)}, \quad (11)$$

which is the fraction of output variance explained by varying $\hat{Z}(k = k_j)$ alone. The total index:

$$S_{T_j} = 1 - \frac{\xi_{\sim j}}{\text{Var}(c_l)}. \quad (12)$$

Here, $\xi_{\sim j} = \text{Var}_{X_{\sim j}}[\mathbb{E}(c_l | X_{\sim j})]$, where $X_{\sim j}$ denotes all inputs except $\hat{Z}(k_j)$.

2.6. Experiments and test cases

We have developed closures for 5 different forced 2D turbulent flows (2 with the β -plane effects; details in Table 1). These cases are commonly used to evaluate the SGS closures of geophysical turbulence [24,48,62] and exhibit distinct behaviors and dynamics, as seen

in snapshots of vorticity fields, ω , in Fig. 1. Training is performed with the objective of achieving an LES enstrophy spectrum close to the target DNS spectrum. We also compare the kinetic energy spectra in Appendix B. Additionally, to test the predictability of SMARL-based LES models for extreme events, we compare the PDFs of the resolved vorticity, $\mathcal{P}(\omega)$. The tails of these PDFs represent rare, extreme events, i.e., significantly large ω with a small probability of occurrence. Note that these vortices resemble the weather system's high- and low-pressure anomalies, which can cause various extreme weather events [63]. LES with 16 to 16,384 $\times$ coarser spatial resolution (in terms of the number of grid points, $(N_{\text{DNS}}/N_{\text{LES}})^2$) and 10 $\times$ larger time steps coupled to the learned closures are then tested with the SMARL-based closures, and their statistics are compared with those of the DNS and LES with classical dynamic Smagorinsky and Leith closures. The computational costs are included in Appendix F.

3. Results and discussion

We evaluated the proposed framework in five test cases, described in detail in Section 2.6. These benchmarks are widely used for atmospheric, oceanic, and planetary flows [20,21,24,48,64,65]. The simulations correspond to a two-dimensional flow with sinusoidal forced turbulence in a periodic domain. This forcing term generates coherent structures of different scales. In addition, some cases contain a Coriolis force β that represents the rotation of the spherical Earth and generates jet streams. The benchmarks are performed at two Reynolds numbers: $\text{Re} = 2 \times 10^4$ (Cases 1–4), a standard regime for benchmarking SGS closures of 2D turbulence [21,48], and $\text{Re} = 3 \times 10^5$ (Case 5), chosen to test the generalization of the learned closures to a 15 $\times$ higher Reynolds number without retraining.

3.1. Prediction of extreme events

We quantify extreme events by estimating the probability density function $\mathcal{P}(\omega)$ of the vorticity field ω over time and space. Positive (cyclonic) and negative (anti-cyclonic) ω in this prototype of atmospheric circulation correspond to low- and high-pressure weather systems, whose large amplitudes (tails) drive various types of extreme events [66]. Fig. 2 shows $\mathcal{P}(\omega)$ computed from cases 1–4 against the prediction performance of RL-Leith, DSmag, and DLeith. In all cases, the RL-Leith model outperforms traditional physics-based dynamic models in terms of matching the DNS PDF. Except for case 4, the RL-Leith PDF matches the DNS within uncertainties, even up at the tails of the distribution, corresponding to low probability or *extreme* events. The lack of extreme events in the DSmag and DLeith predictions is attributed to the excessive diffusion from the positive clipping essential for stability [20]. This is consistent with Fig. 3 showing the excessive interscale enstrophy transfer predicted by DSmag and DLeith for all cases. Quantitative confidence intervals for the higher-order moments and the 3σ tail probabilities are reported in Appendix D. These results show that RL-Leith substantially improves the frequency of extreme-vorticity events relative to DSmag and DLeith. We note, however, that RL-Leith still underestimates the 3σ tail probability in Cases 1, 3 and 4, and that none of the closures reproduces the skewness of the DNS vorticity distribution; see Appendix D. Additional *a-posteriori* analysis, including the turbulent kinetic energy spectra, is included in Appendix B.

3.2. Enstrophy transfer between scales

A main criterion for a functional eddy-viscosity LES model is the correct prediction of the total energy/enstrophy transfer between resolved and subgrid scales [50]. Fig. 3 shows the interscale enstrophy transfer ($\langle P_z \rangle = \langle \Pi \omega \rangle$) of RL-Leith, DSmag, and DLeith compared with the reference $\langle P_z \rangle$ obtained from filtering the DNS results. Here, $P_z > 0$ denotes forward enstrophy transfer and SGS dissipation, whereas

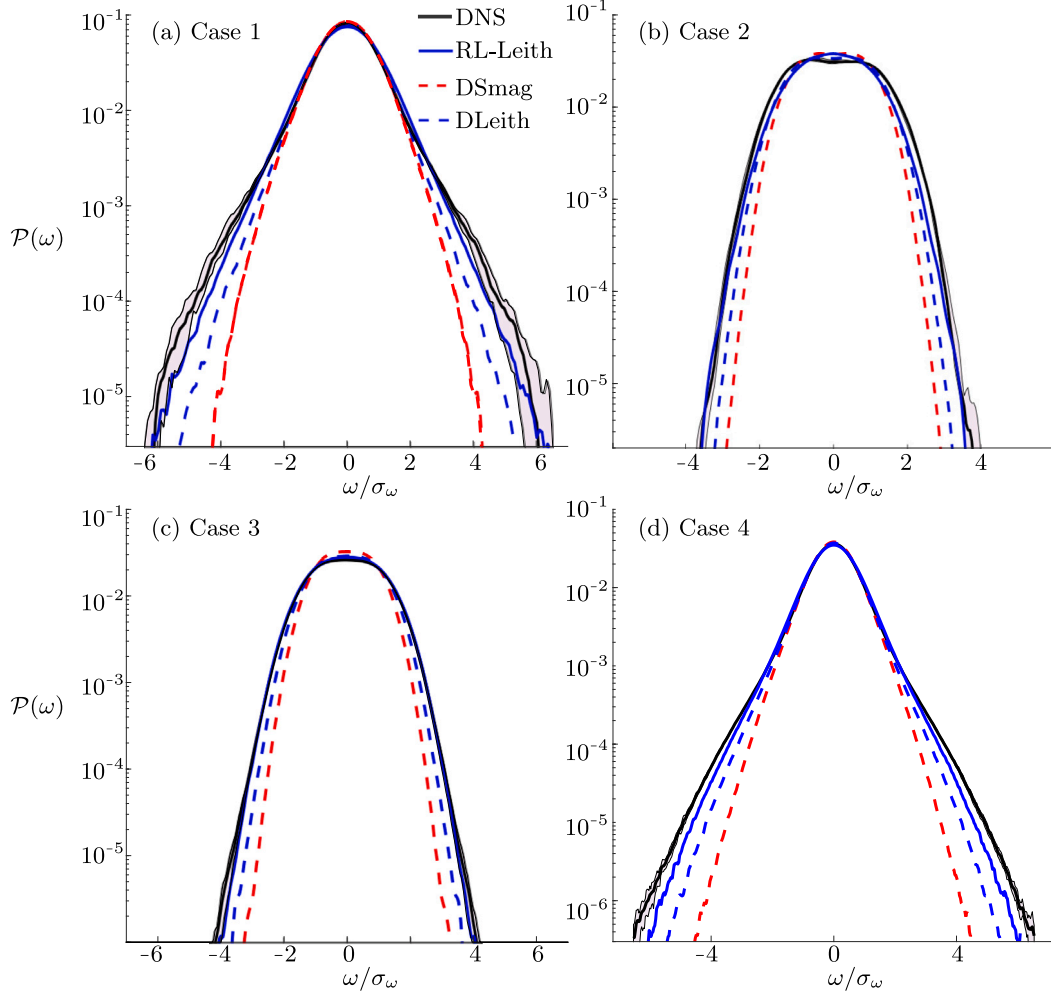


Fig. 2. *A-posteriori* testing results in terms of the PDF of vorticity ($\mathcal{P}(\omega)$). The x-axis is normalized by the standard deviation of vorticity (σ_ω) computed from DNS of each case. The shading area represents the uncertainty (25–75 quartiles) of the DNS PDF.

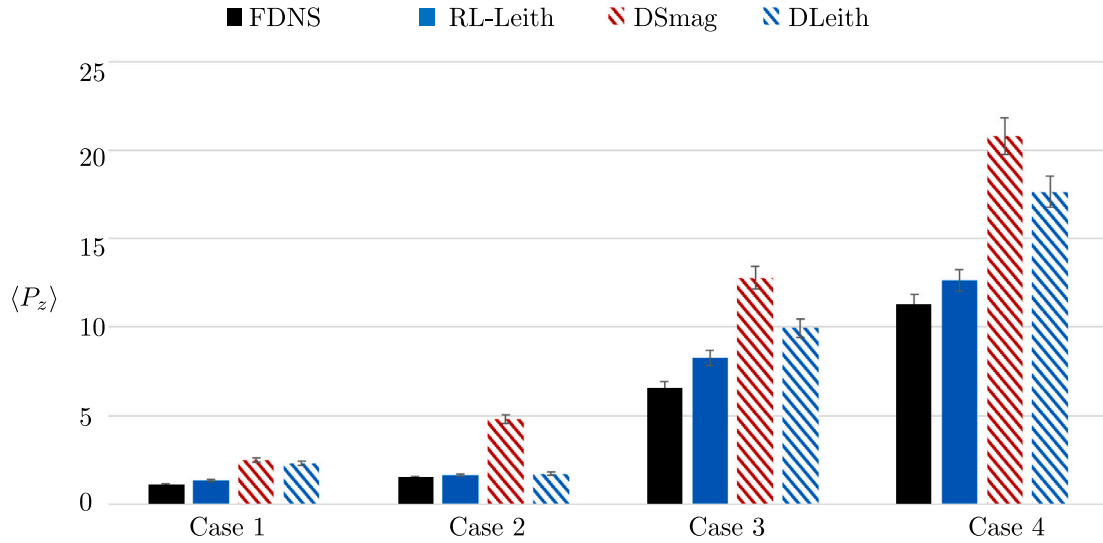


Fig. 3. Total enstrophy diffusion $\langle P_z \rangle$ obtained from the FDNS, RL-Leith, DSmag and DLeith for the four cases. The error bars indicate 95% confidence intervals estimated using temporal block bootstrap.

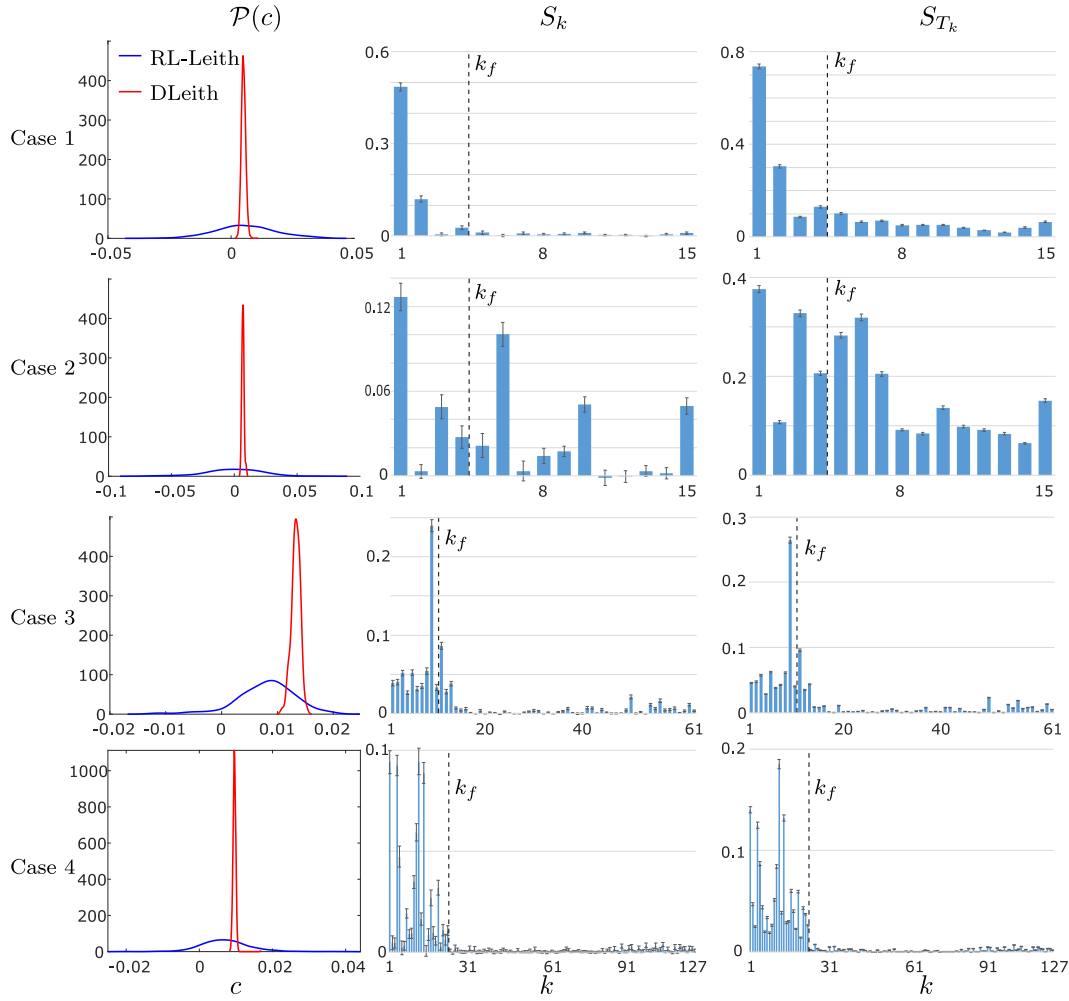


Fig. 4. First column: PDF of model coefficients c ; second column: S_k ; and third column: S_{T_k} . The error bars on the Sobol indices indicate 95% confidence level. The Nyquist (or cut-off) wavenumber k_c is the maximum wavenumber (i.e., $k_c = 16$ in Case 1 and 2, $k_c = 61$ in Case 3, and $k_c = 127$ in Case 4). The dashed lines in second and third columns mark the forcing wavenumber k_f .

$P_z < 0$ denotes backscattering. The RL-Leith model outperforms DSmag in all four cases, and DLeith in Cases 1, 3 and 4, where the 95% confidence intervals of the two models do not overlap; in Case 2 the RL-Leith and DLeith predictions are statistically indistinguishable. Overall, RL-Leith reduces the over-prediction of $\langle P_z \rangle$ from 52%–216% (DSmag and DLeith) to 8%–26%, although it still over-predicts the interscale transfer relative to the FDNS reference. This is likely due to the ability of RL-Leith to capture backscattering (transfer from subgrid to resolved scales), resulting in less diffusion than DSmag and DLeith. We also observe that DLeith outperforms DSmag in all 4 cases, demonstrating the benefits of using the Leith form of the eddy-viscosity model rather than the Smagorinsky form in LES of 2D turbulence, where capturing enstrophy interscale transfer is essential. The advantage of DLeith over DSmag in terms of capturing the interscale enstrophy transfer is also reflected on *a-posteriori* analysis, see Fig. 2. The probability density functions ($P(P_z)$) reported in Appendix C further demonstrate that RL-Leith substantially improves the representation of local interscale enstrophy transfer. In contrast to the positive-clipped dynamic closures, RL-Leith captures both forward transfer and backscattering and provides better agreement with the broad FDNS distribution, including its intermittent tails.

3.3. Interpretability of the SMARL model

To understand the superior performance of RL-Leith over the classical SGS models, we compare the distribution of Leith coefficients c_l

obtained from RL-Leith and DLeith. The first column of Fig. 4 shows the PDF of the Leith model coefficients c_l for the four cases. In all cases, the coefficients obtained with the DLeith model are concentrated around some positive values (rare negative values have been removed with positive clipping, which is essential for the LES stability with DLeith and DSmag [20,22]). This leads to excessive diffusion, also confirmed by Figs. 2 and 3. In contrast, the Leith coefficients learned from RL-Leith spread across a much wider range of values than DLeith, and include negative values (LES with Leith-RL remains stable despite these negative viscosities). This shows that the RL-Leith model captures both the forward/diffusive and backscatter inter-scale energy/enstrophy transfer. Similarly to DLeith, the majority of the coefficients predicted by RL-Leith is positive, showing that the RL-Leith model is more likely to be diffusive in general. However, the highest probability density of the coefficients obtained with RL-Leith occurs closer to 0, especially for Cases 3 and 4, indicating that the RL-Leith is less diffusive than DLeith.

The second and third columns of Fig. 4 show the sensitivity of the model to its inputs. This is shown through the first-order and total Sobol indices of the model policy (see Appendix E for details). It decomposes the variance of the model output into contributions from: (1) each individual input (first-order Sobol index S_k), and (2) interaction among inputs (total Sobol index $S_{T,k}$). In all cases, the model output is more sensitive to the low-wavenumber entries of the enstrophy spectrum. This observation is consistent with the fact that most of the

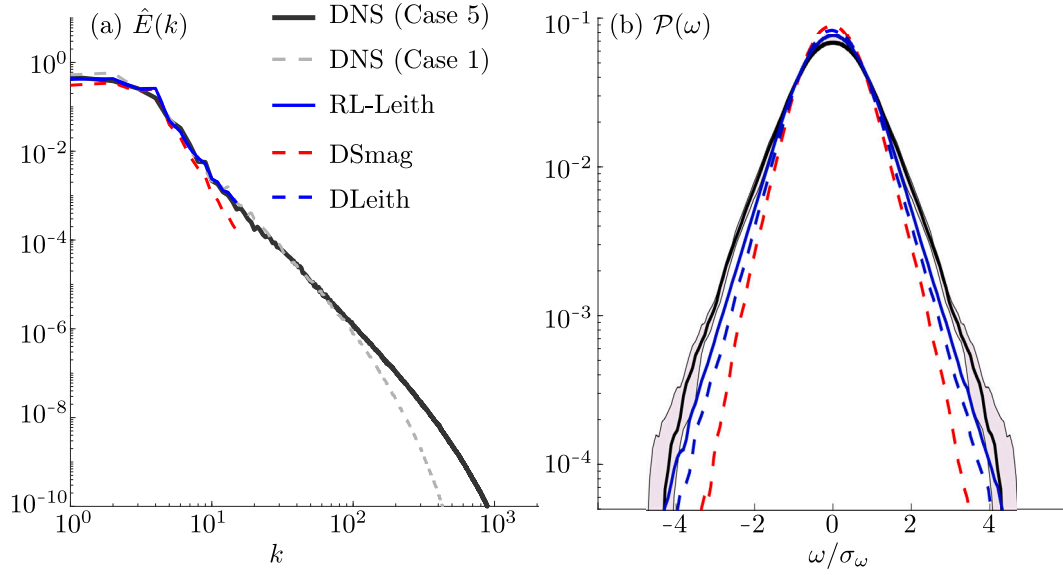


Fig. 5. Generalization testing of SMARL to higher ($15 \times Re$) turbulence (SMARL trained on Case 1 and tested on Case 5). (a) shows the turbulent kinetic energy spectra, $\hat{E}(k)$, and (b) shows the PDF, $\mathcal{P}(\omega)$. In (b), the x-axis is normalized by the standard deviation of vorticity (σ_ω) computed from DNS. The shading area represents the uncertainty (25–75 quartiles) of the DNS PDF.

kinetic energy is contained in the large scales, corresponding to low wavenumbers k . For example, the ratio of turbulent kinetic energy $\langle \hat{E}(k < k_f) \rangle_k / \langle \hat{E}(k) \rangle_k = 92.6\%$ for Case 1, 94.5% for Case 2, 92.3% for Case 3, and 93.4% for Case 4. In 2D turbulence, this region also coincides with the inverse energy cascade region, $k < k_f$, where k_f is the forcing wavenumber of the 2D turbulence systems at which energy is injected into the flow (see Section 2). The next significant region is near the Nyquist wavenumber k_c , i.e. the maximum wavenumber in the LES simulations. This region is responsible for interscale energy and enstrophy transfer between the resolved and unresolved scales. Therefore, changes of $\hat{Z}$ at this wavenumber immediately trigger a change in the closure coefficient c_l . The Sobol indices suggest that the enstrophy contained in the middle wavenumber range does not have a significant impact on the learned coefficients c_l , and thus the learned viscosity.

3.4. Generalization to higher Reynolds numbers

The generalization capability of data-driven SGS models across different flow regimes (e.g., higher Re turbulent flows) is essential for their practical use. However, many deep-learning-based SGS models do not generalize well and often require extra data or training efforts, such as transfer learning [20,25] to be successful. Training data for high- Re turbulent flows are especially difficult to obtain. For example, Case 5 has $15\times$ higher Re than Case 1, and therefore DNS to resolve Case 5 requires $16\times$ higher spatial resolution compared to Case 1 and correspondingly higher temporal resolution due to Courant–Friedrichs–Lewy condition. In Fig. 5 we show that the RL-Leith closure trained on Case 1 can be directly used for Case 5 without any new data from Case 5 or an additional training process. It can be observed that the RL-Leith trained on Case 1 generalizes successfully to Case 5. Similar to the testing for Case 1, the RL-Leith model outperforms DSmag in terms of matching the DNS kinetic energy spectra, $\hat{E}(k)$, and it outperforms both physics-based models (DSmag and DLeith) in matching the DNS $\mathcal{P}(\omega)$ for Case 5 down to the tails (extreme events). The generalization capability of SMARL-based SGS models to higher Re turbulence can be explained by the state-action map that is trained during the RL online learning. As the state, $s'(t)$, is the enstrophy spectrum, $\hat{Z}(k, t)$, the difference between $Re = 20,000$ (Case 1) and $Re = 300,000$ (Case 5) lies only in the smallest scales that can only be resolved in DNS. When the LES uses an aggressive (small) cut-off wavenumber (e.g., here for both Case

1 and 5, $k_c = 16$), the cut-off $\hat{Z}_{\text{DNS}}(k, t)$ is very similar between Case 1 and 5, see Fig. 5 for the comparison between the energy spectrum, $\hat{E}(k)$, of Case 1 and 5.

4. Conclusions

We have introduced a data-efficient, SMARL-based framework to learn LES closures for geophysical turbulence with broad potential applications to climate modeling. This framework is superior to existing AI-based closure modeling approaches as it works with: (i) sparse amounts of training data that are insufficient for supervised learning based on DNN [48] and (ii) low LES resolutions inaccessible to closures identified with equation discovery [21]. Furthermore, the framework does not need a differentiable numerical solver and implicitly addresses model structural errors [40,67].

We show that these closures enable LES to produce statistics, including energy spectra, PDF, and most importantly, tails of the PDFs, that closely match those of the far more expensive DNS, with the frequency of extreme vorticity events predicted to within a factor of 2.6 of the reference, although the most extreme tails remain underestimated (see Appendix D). Moreover, LES with SMARL-based closures significantly outperform LES with classical physics-based closures in capturing extreme events, which are often diffusive to ensure stability but then underestimate extremes. With a small number of samples from DNS, SMARL develops closures that substantially improve the statistics of such extreme events, suggesting that both diffusion and backscattering, i.e., the two-way interaction of the resolved scales and SGS, are represented. We emphasize, however, that while the sign and frequency of backscatter are captured, in contrast to the positively clipped dynamic closures that exclude it altogether, its most extreme magnitudes are underestimated (see Appendix C), and the asymmetry of the vorticity distribution is not reproduced (see Appendix D). We emphasize that the LES cases investigated here have 160 to 163,840 $\times$ spatio-temporally coarser resolution compared to their DNS. LES+SMARL is stable and can easily run thousands of times longer than the lengths of the training set and training horizon.

We provide an interpretation of the SMARL-based closure by analyzing the state-action map learned using SMARL with the Sobol indices. Their distribution across wavenumber, k , in the enstrophy spectrum indicates that large-scale structures that contain most of the kinetic energy make the largest contribution to the prediction of the closure's

coefficient c_l . The next significant contribution stems from the high-wavenumber region near the cut-off wavenumber, k_c . This can be interpreted as the directly responsible region of interscale energy and enstrophy transfer.

We have tested the generalization of the SMARL-based closure by testing a pre-trained DNN to a different flow regime where the Re is increased by 15 $\times$. LES test shows that the SMARL-based closure, albeit trained at a lower Re -number flow, performs better than DLeith and DSmag in terms of matching the kinetic energy spectrum and the PDF of vorticity, $P(\omega)$, even at the tails of the distribution, corresponding to extreme vorticity events.

We conclude by noting limitations of the present study and directions for future work. Each closure was trained for a given forcing scheme; while we demonstrated generalization to a 15 $\times$ higher Reynolds number, the transfer of a policy across forcing schemes, as well as to decaying or otherwise statistically unsteady flows, remains to be assessed. For statistically unsteady flows, the main required modification lies in the reward, which would compare against time-resolved (e.g., phase- or ensemble-averaged) reference statistics instead of the stationary spectrum used here. Furthermore, the framework is not restricted to the eddy-viscosity ansatz, which only defines the action space of the agents: natural extensions include spatially varying coefficients driven by local states, actions parameterizing more general structural forms of the closure, and the direct prediction of the SGS fluxes by the agents, with the latter posing, however, challenges for the stability of the LES.

In summary, we demonstrate a SMARL-based framework to develop novel, accurate, stable, and generalizable SGS closure for canonical geophysical turbulence, with potential applications to larger scale problems such as GCMs and Earth system models.

CRedit authorship contribution statement

Yifei Guan: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Lucas Amoudruz:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Sergey Litvinov:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Karan Jakhar:** Methodology. **Rambod Mojgani:** Software, Methodology, Investigation. **Petros Koumoutsakos:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Pedram Hassanzadeh:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Nomenclature

See [Tables A.2](#) and [A.3](#).

Table A.2

List of the main symbols used in this paper.

<i>Flow and simulations</i>	
ω, ψ	vorticity and stream function
$\mathcal{N}$	nonlinear advection operator
Re	Reynolds number
α	linear (Rayleigh) drag coefficient
β	meridional gradient of the Coriolis parameter
f, k_f	forcing and forcing wavenumber
L	domain length
N_{DNS}, N_{LES}	number of grid points per direction
$\Delta t_{DNS}, \Delta t_{LES}$	DNS and LES time steps
$\bar{\cdot}$	filtered quantity
k, k_c	wavenumber and LES cut-off wavenumber
$\hat{Z}(k, t), \hat{E}(k)$	enstrophy and kinetic energy spectra
σ_ω	standard deviation of vorticity
$P(\omega)$	probability density function of vorticity
$\tau^{SGS}, \bar{S}_{ij}, \nu_e$	subgrid stress, filtered strain-rate tensor, eddy viscosity
c_s, c_l	Smagorinsky and Leith model coefficients
Π	subgrid-scale term in the filtered equations
$\langle P_z \rangle$	interscale enstrophy transfer
<i>Reinforcement learning</i>	
$N, n_{A,x} \times n_{A,y}$	number and arrangement of agents
$s'(t), s^{(i)}(t)$	global state and state of agent i
$a^{(i)}(t)$	action of agent i
$r(t)$	reward
π^θ, μ_t	policy and behavior policy
θ	neural network weights
V^θ	value function
γ	discount factor
ρ_k	importance weight
c_{max}, β_{RF}	ReF-ER hyper-parameters
Δt_{RL}	actuation interval
<i>High-order statistics</i>	
S_m, K_m	skewness and kurtosis
$p^\pm$	two-sided tail probability
<i>Sensitivity analysis</i>	
S_j, S_{T_j}	first-order and total Sobol indices
ξ_m	partial variance

Table A.3

List of abbreviations.

DNS	direct numerical simulation
FDNS	filtered direct numerical simulation
LES	large-eddy simulation
SGS	sub-grid scale
EKI	ensemble Kalman inversion
RL	reinforcement learning
SMARL	scientific multi-agent reinforcement learning
DNN	deep neural network
GCM	global climate model
PDF	probability density function
DSmag, DLeith	dynamic Smagorinsky and dynamic Leith closures
RL-Smag, RL-Leith	SMARL-based Smagorinsky and Leith closures

Appendix B. A-posteriori analysis in turbulent kinetic energy spectra

In this section, we compare the turbulent kinetic energy spectra, $\hat{E}(k)$, of the LES with SMARL-based (RL-Smag and RL-Leith) and traditional dynamic Smag and Leith (DSmag and DLeith) models with the ones obtained from DNS (shown in [Fig. B.6](#)). The SMARL-based models match the DNS/FDNS spectra over the whole spectrum $k \in [0, k_c]$ across all 4 cases, outperforming both the DSmag and DLeith models.

Appendix C. Probability density functions of interscale enstrophy transfer P_z

In this section, we compare the PDF of interscale enstrophy transfer P_z , of the LES with SMARL-based (RL-Leith) and traditional dynamic

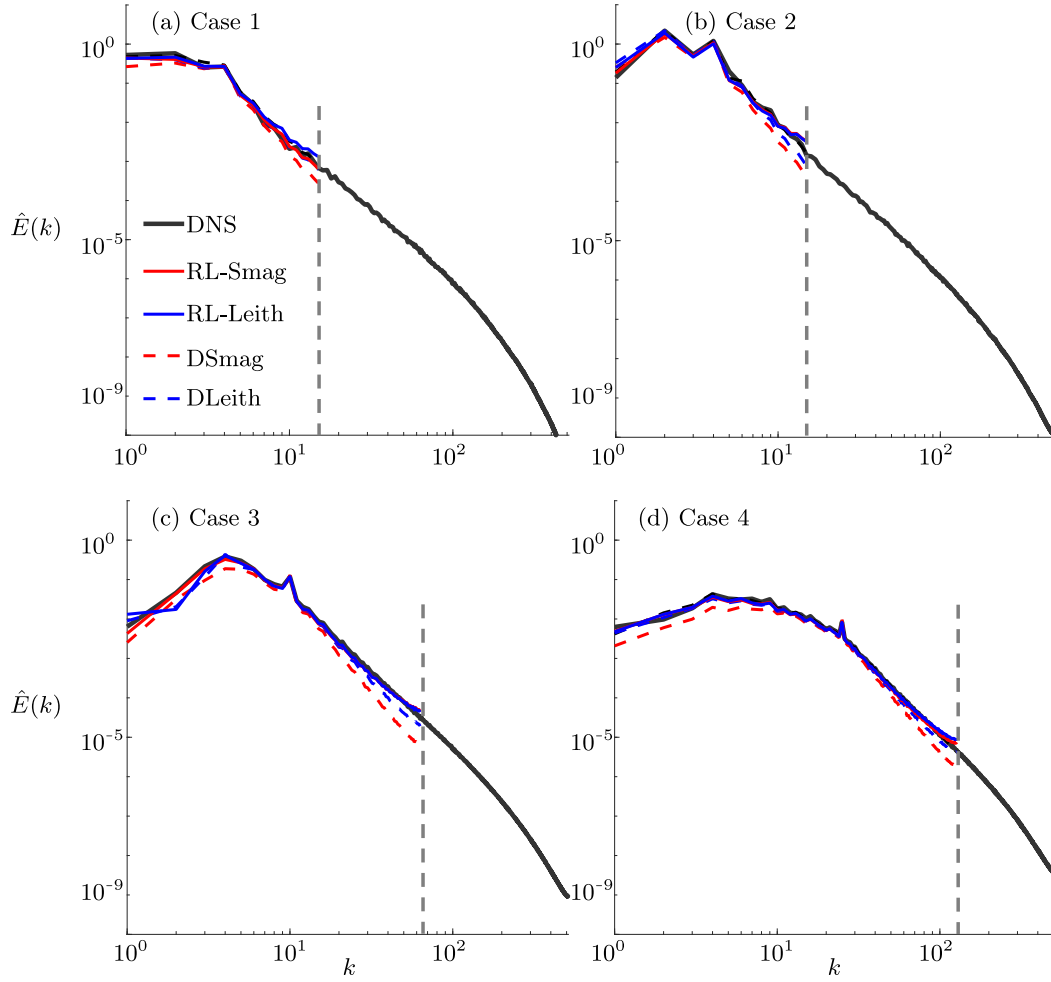


Fig. B.6. *A-posteriori* testing results in terms of turbulent kinetic energy spectra, $\hat{E}(k)$. The gray dashed lines mark the cut-off wavenumber, k_c . The FDNS spectra are the same as the DNS spectra up to k_c .

Smag and Leith (DSmag and DLeith) models with the ones obtained from DNS (FDNS), as shown in Fig. C.7. The FDNS distribution exhibits broad positive and negative tails, revealing strongly intermittent and bidirectional interscale enstrophy transfer. Backscattering occurs with a frequency comparable to forward-transfer events, indicating that backscattering represents an essential component of the local SGS dynamics rather than a rare correction. The positive-clipped DSmag and DLeith models are intrinsically unable to reproduce the negative-transfer branch and yield distributions concentrated almost entirely on positive transfer. RL-Leith substantially improves the qualitative agreement by recovering both forward-transfer and backscattering events. However, its PDF remains considerably narrower than the FDNS distribution, particularly in the extreme tails, showing that the learned closure still underestimates the intermittency and magnitude of SGS interscale transfer.

Appendix D. Statistical robustness of the vorticity tails

To quantify the statistical robustness of the vorticity tails, we calculate the skewness, kurtosis, and probability of extreme vorticity using the temporally resolved two-dimensional vorticity snapshots. Let $\omega_m^{(n)}(x_i, y_j)$ denote the vorticity predicted by model m (e.g., FDNS, RL-Leith, etc.) at the grid point (x_i, y_j) in snapshot n , where $i = 1, \dots, N_x$, $j = 1, \dots, N_y$, and $n = 1, \dots, N_t$. Here, N_x and N_y are the numbers of grid points in the x and y directions, respectively, and N_t is the number of snapshots.

The space-time average of a quantity f that depends on $\omega_m^{(n)}(x_i, y_j)$ is defined as

$$\langle f(\omega_m) \rangle = \frac{1}{N_t N_x N_y} \sum_{n=1}^{N_t} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} f(\omega_m^{(n)}(x_i, y_j)). \quad (\text{D.1})$$

The model-dependent mean and variance are

$$\mu_m = \langle \omega_m \rangle, \quad \sigma_m^2 = \langle (\omega_m - \mu_m)^2 \rangle. \quad (\text{D.2})$$

The skewness and Pearson kurtosis are calculated as

$$S_m = \frac{\langle (\omega_m - \mu_m)^3 \rangle}{\sigma_m^3}, \quad K_m = \frac{\langle (\omega_m - \mu_m)^4 \rangle}{\sigma_m^4}. \quad (\text{D.3})$$

With this definition, a Gaussian distribution has $S_m = 0$ and $K_m = 3$. To compare the far tails produced by the different closures directly, we use a constant threshold based on the corresponding FDNS. The two-sided tail probability is defined as

$$p_{3\sigma, m}^{\pm} = \frac{1}{N_t N_x N_y} \sum_{n=1}^{N_t} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \mathbb{I} \left[\left| \omega_m^{(n)}(x_i, y_j) - \mu_{\text{FDNS}} \right| > 3\sigma_{\text{FDNS}} \right], \quad (\text{D.4})$$

where $\mathbb{I}[\cdot]$ is the indicator function, and μ_{FDNS} and σ_{FDNS} are the mean and standard deviation of the corresponding FDNS vorticity field. The same FDNS-based threshold is applied to all LES closures for a specific case. Therefore, differences in $p_{3\sigma, m}^{\pm}$ represent differences in the predicted occurrence rate of extreme vorticity rather than differences caused by model-dependent normalization.

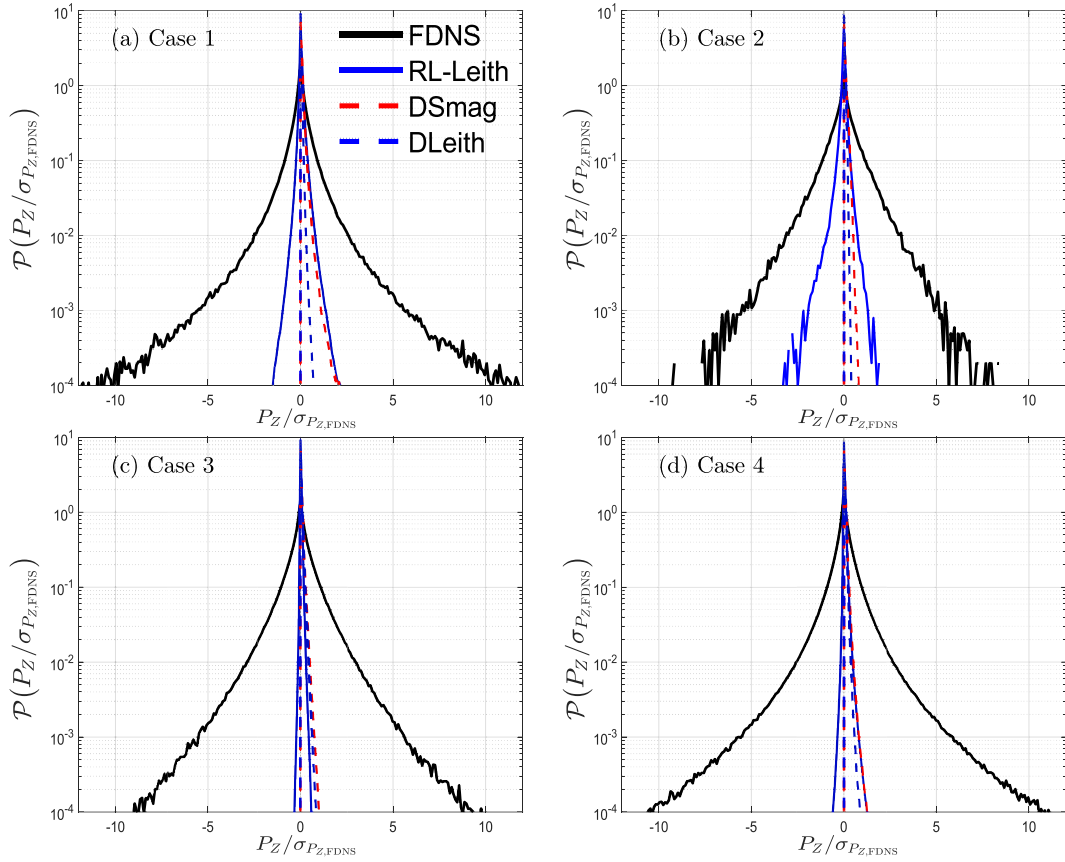


Fig. C.7. Probability density functions of the normalized interscale entrophy transfer ($P_z / \sigma_{P_z, \text{FDNS}}$) for four cases. FDNS exhibits broad positive and negative tails with comparable forward-transfer and backscattering probabilities. RL-Leith captures the bidirectional transfer more accurately than the positive-clipped DSmag and DLeith models. The PDFs of DSmag and DLeith models are non-zero only in the positive P_z region (pure diffusion) due to positive clipping.

Table D.4

Higher-order vorticity statistics and 95% temporal block-bootstrap confidence intervals. The kurtosis is the Pearson kurtosis, for which a Gaussian distribution has $K = 3$. The two-sided tail probability uses the FDNS threshold $|\omega - \mu_{\text{FDNS}}| > 3\sigma_{\text{FDNS}}$. A reported zero indicates that no event exceeding the specified threshold was observed in the available snapshots and should not be interpreted as a theoretical zero probability.

Case	Model	Skewness S	Kurtosis K	$p_{3\sigma}^{\pm}$
1	FDNS	0.082 [−0.096, 0.245]	5.136 [4.727, 5.553]	1.32×10^{-2} [9.89×10^{-3} , 1.66×10^{-2}]
1	RL-Leith	0.003 [−0.007, 0.012]	4.124 [4.079, 4.172]	8.65×10^{-3} [8.37×10^{-3} , 8.93×10^{-3}]
1	DSmag	0.034 [−0.004, 0.069]	3.521 [3.425, 3.632]	1.75×10^{-3} [1.36×10^{-3} , 2.21×10^{-3}]
1	DLeith	0.003 [−0.052, 0.063]	3.963 [3.818, 4.139]	4.59×10^{-3} [3.82×10^{-3} , 5.43×10^{-3}]
2	FDNS	0.087 [0.067, 0.106]	2.339 [2.302, 2.372]	2.25×10^{-4} [6.84×10^{-5} , 4.79×10^{-4}]
2	RL-Leith	0.016 [−0.011, 0.041]	2.582 [2.558, 2.605]	3.22×10^{-4} [2.25×10^{-4} , 4.20×10^{-4}]
2	DSmag	0.001 [−0.003, 0.004]	2.170 [2.147, 2.194]	0
2	DLeith	−0.001 [−0.026, 0.024]	2.409 [2.402, 2.416]	7.81×10^{-6} [1.95×10^{-6} , 1.46×10^{-5}]
3	FDNS	0.038 [0.024, 0.052]	2.556 [2.534, 2.576]	8.29×10^{-4} [6.58×10^{-4} , 1.01×10^{-3}]
3	RL-Leith	−0.008 [−0.018, 0.003]	2.518 [2.499, 2.536]	3.26×10^{-4} [2.44×10^{-4} , 4.12×10^{-4}]
3	DSmag	−0.002 [−0.013, 0.009]	2.459 [2.437, 2.480]	1.34×10^{-5} [0, 3.72×10^{-5}]
3	DLeith	−0.002 [−0.012, 0.008]	2.467 [2.453, 2.482]	1.15×10^{-4} [8.30×10^{-5} , 1.51×10^{-4}]
4	FDNS	−0.169 [−0.181, −0.158]	5.575 [5.489, 5.664]	1.31×10^{-2} [1.28×10^{-2} , 1.35×10^{-2}]
4	RL-Leith	0.000 [−0.004, 0.003]	4.883 [4.868, 4.898]	1.13×10^{-2} [1.12×10^{-2} , 1.13×10^{-2}]
4	DSmag	0.002 [−0.004, 0.009]	3.961 [3.932, 3.991]	3.39×10^{-3} [3.29×10^{-3} , 3.47×10^{-3}]
4	DLeith	−0.002 [−0.020, 0.011]	4.681 [4.649, 4.720]	7.89×10^{-3} [7.77×10^{-3} , 8.02×10^{-3}]

The 95% confidence intervals are estimated using a temporal block bootstrap. Each trajectory is divided into 20 contiguous temporal blocks, and these blocks are resampled with replacement 5000 times. All $N_x N_y$ spatial grid points within a selected temporal block are retained together. This procedure preserves the instantaneous two-dimensional spatial correlations and partially accounts for temporal correlations between consecutive snapshots (see Table D.4).

The confidence intervals confirm that the non-Gaussian character of the FDNS vorticity distributions is statistically robust. Cases 1 and

4 exhibit super-Gaussian distributions with pronounced heavy tails, whereas Cases 2 and 3 are sub-Gaussian, with kurtosis significantly below three. Therefore, the vorticity PDFs in Cases 2 and 3 should not be interpreted as Gaussian despite their comparatively light tails.

Among the three LES closures, RL-Leith gives the closest two-sided 3σ tail probability to FDNS in all four cases. It also provides the closest kurtosis in Cases 1, 3, and 4, whereas DLeith gives the closest kurtosis in Case 2. These results indicate that RL-Leith improves the overall

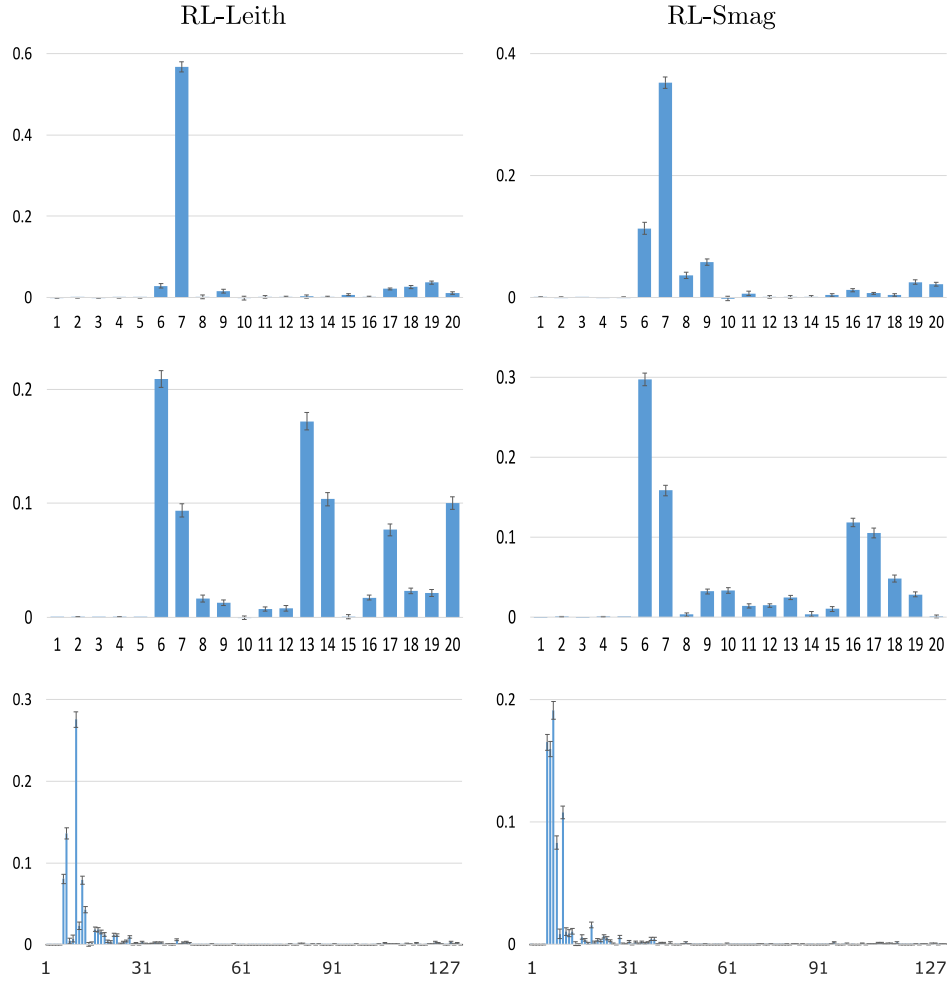


Fig. E.8. First-order Sobol indices of the control policies trained with RL-Leith and RL-Smag formulations for all (local + global) states in Cases 1, 2, and 4 (from top to bottom).

frequency and magnitude of extreme vorticity fluctuations, although it still underestimates the FDNS tail probability in Cases 1, 3, and 4.

In contrast, the skewness is not generally reproduced by the LES closures. In Case 1, the FDNS confidence interval includes zero, and the nearly zero LES skewness values are therefore statistically consistent with the FDNS result. In Cases 2 and 3, however, FDNS exhibits small but statistically significant positive skewness, whereas all LES closures produce nearly symmetric distributions. The discrepancy is most pronounced in Case 4, where the significant negative FDNS skewness is absent from all three LES results. Thus, while RL-Leith improves the thickness of the vorticity tails, it does not fully reproduce their directional asymmetry.

Appendix E. Sensitivity analysis of the policy

In this section, we investigate which states are responsible for the most change in the output of the policy. To do so, we train the RL agents in cases 1, 2, 3, and 4 (see Section 2.6) with both the Smagorinsky and Leith models. The input states contain the five nonzero local invariants of the velocity gradient and velocity Hessian, followed by the logarithm of the enstrophy spectrum. We then compute the first-order Sobol indices of the policy for each input state, using the SALib package [68].

Fig. E.8 shows the first-order Sobol indices for all three cases with the RL-Leith and RL-Smagorinsky models, respectively. We observe that in all cases, the first five states, corresponding to the local invariants of the velocity gradient (2 local invariants in 2D turbulence) and velocity Hessian (3 local invariants in 2D turbulence) [41], have a nearly zero

first-order Sobol index. This means that the policy does not depend on these states, and we have thus chosen to remove them from the input of the policies in this study. We also note that in most cases, input states that correspond to low frequencies of the enstrophy spectrum have a dominant first-order Sobol index, indicating that high frequencies do not play a major role in the control of the coefficients c_s and c_l .

The effects of local states on the *a-posteriori* results are shown in Fig. E.9. In terms of predicting the probability density function of vorticity, $P(\omega)$, the RL models with local states perform similarly to those with global states only (see Fig. 2). Another note is that Smag and Leith perform similarly with RL optimized coefficients, which is reasonable due to the equivalence of both models in 2D once their coefficients are optimized [40,69].

Appendix F. Computational costs

Table F.5 reports the wall-clock cost of SMARL training on a Jetstream2 virtual machine using 16 virtual CPU cores based on an AMD EPYC-Milan processor. Consistent with the training horizon specified in Table 1, each case is trained for 10^3 RL control intervals (Δt_{RL}). Because the policy is updated once every 10 LES time steps, this training horizon corresponds to 10^4 LES time steps.

The total training cost consists of three components: LES integration, RL-related operations, and serialization. The RL-related cost includes policy evaluation and policy optimization, whereas serialization includes data transfer and other bookkeeping operations. The small

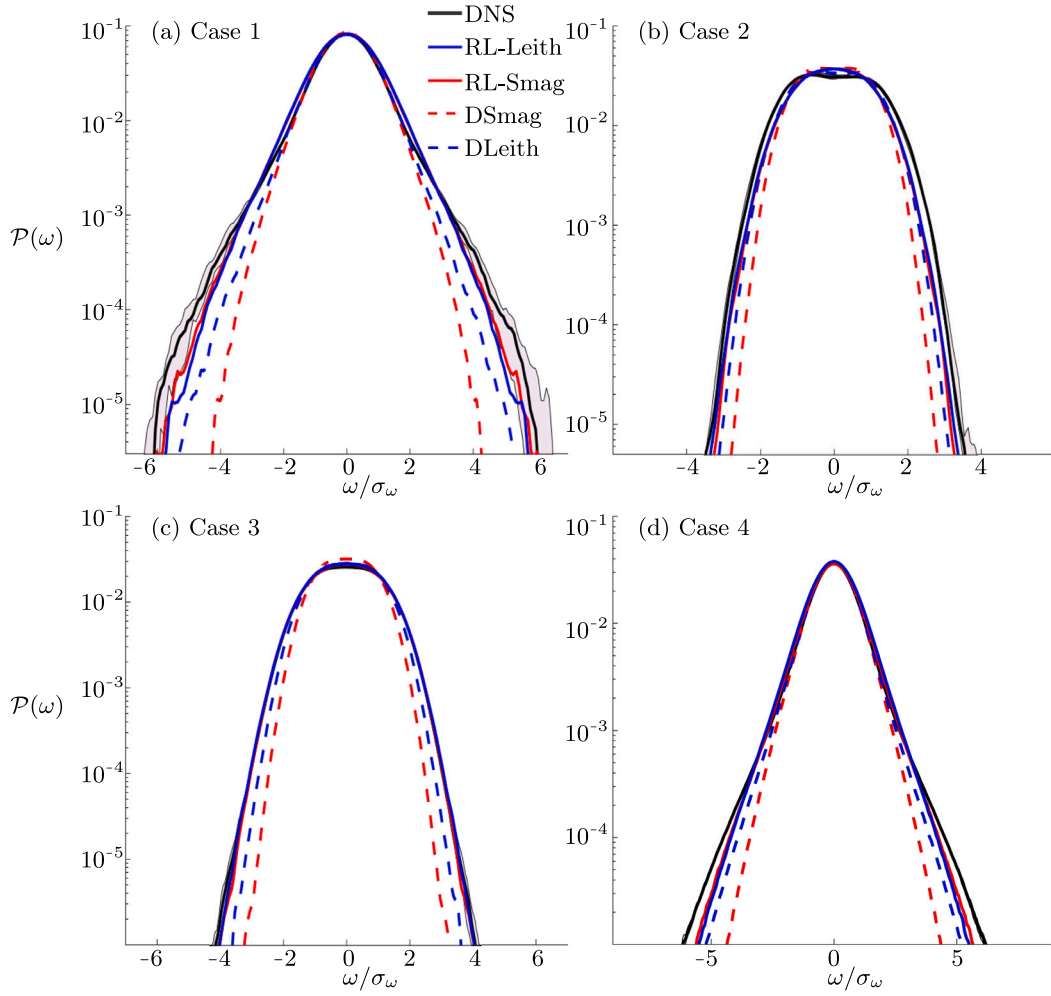


Fig. E.9. *A-posteriori* testing results in terms of probability density function of vorticity ($P(\omega)$). The x-axis is normalized by the standard deviation of vorticity (σ_ω) computed from DNS of each case. The shading area represents the uncertainty (25–75 quartiles) of the DNS PDF. Here, the RL models take both local states (5 invariants of velocity gradient and velocity Hessian tensors) and global states (logarithm of enstrophy spectrum) as inputs.

Table F.5

Computational cost of SMARL training on a Jetstream2 virtual machine using 16 virtual CPU cores based on an AMD EPYC-Milan processor. Each case uses 16 agents and one concurrent LES environment. The training horizon is 10^3 RL control intervals, corresponding to 10^4 LES time steps. The long-LES cost corresponds to the testing horizon of 10^6 RL control intervals, or 10^7 LES time steps.

Case	N_{LES}	Training-cost breakdown (s)				Long-LES cost (h)	Training/LES ($\times 10^{-3}$)
		LES	RL	Serialization	Total		
1	32	11.8	21.4	1.5	35.3	3.61	2.71
2	32	10.4	23.0	1.1	34.6	3.14	3.06
3	128	69.0	21.7	2.4	93.6	19.41	1.34
4	256	219.5	24.8	4.4	249.3	61.24	1.13

difference between the sum of the listed components and the total cost is due to scheduling and other runtime overhead.

At $N_{\text{LES}} = 32$, policy evaluation and optimization constitute the largest fraction of the training cost. As the LES resolution increases to $N_{\text{LES}} = 128$ and 256, the LES integration becomes the dominant component, while the RL-related cost remains approximately constant. The complete 10^3 -update training horizon requires approximately 35–249 s across the four cases. This is only approximately 0.11%–0.31% of the cost of one long *a-posteriori* LES over the testing horizon. Therefore, the computational overhead associated with SMARL training is small relative to the subsequent long-term LES simulations.

Data availability

No data was used for the research described in the article.

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